Prime-Target Defect Forms: Arithmetic Involutions, a Bombieri–Vinogradov Small-Divisor Ledger, and Exact Fixed-CRT Barriers

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Abstract

Finite prime-sieve dynamics detects whether a target $mn + h$ avoids prescribed local divisors, but a prime target requires the von Mangoldt weight $\Lambda(mn + h)$. We record this transition exactly without treating it as a parity-breaking theorem. The truncated divisor identity

$$\Lambda_{\le R}(N) = -\sum_{\substack{d|N \\ d\le R}} \mu(d)\log d$$

places every finite-resolution target-prime form in the same congruence algebra as the centered sieve kernels of the preceding papers. For $(d, h) = 1$, the congruence $xy \equiv -h \pmod d$ is the graph of the measure-preserving involution $x \mapsto -hx^{-1}$ on $(\mathbb{Z}/d\mathbb{Z})^\times$. We give its exact character expansion and spectrum. The actual d-layer has norm $|\mu(d)| \log d/\varphi(d)$. The prime-modulus layer norms have cumulative mass $\log R + O_h(1)$, so an argument based only on their absolute sum cannot inherit the summable conductor majorant used for the rough kernel. Since the layers act on different finite spaces, this is not the norm divergence of a single operator.

For the smoothed fixed-shift correlation, expanding the first von Mangoldt factor gives an exact divisor decomposition:

$$\mathcal{C}_h(X; W) = \sum_n \Lambda(n)\Lambda(n + h)W(n/X).$$

Classical Bombieri–Vinogradov theory evaluates the signed sub-sum with coprime divisors $d \le X^{1/2}(\log X)^{-B}$. The main term of this truncation is

$$\mathfrak{S}(h)X \int_0^\infty W(t)\,dt + o(X),$$

where $\mathfrak{S}(h)$ is the Hardy–Littlewood pair singular series. The full correlation differs from that expression by an explicit complementary large-divisor Möbius–prime tail, for which no estimate is obtained. Consequently the conjectural asymptotic is equivalent to cancellation of that tail; the equivalence is an exact ledger, not an independently easier theorem.

Finally, we prove two finite-information barriers. Every nonnegative observable that is periodic in the two factor coordinates and is a pointwise minorant of $\mathbf{1}_{\mathbb{P}}(mn + h)$ on all sufficiently large pairs is identically zero. We also construct two nonnegative point masses with identical pushforward modulo any fixed odd Q, hence identical finite CRT and character data, as well as identical centered-kernel data at $\mathrm{rad}(Q)$, while one target is prime and the other is a square with the opposite Liouville sign. An exact integer certificate is supplied for $Q = 5 \cdot 7 \cdot 11$. These statements concern fixed finite local information; they do not rule out growing-resolution, dispersion, or Type II methods. No twin-prime lower bound is proved.

Keywords: prime-target defect; von Mangoldt function; Bombieri–Vinogradov theorem; arithmetic involution; sieve parity; finite CRT dynamics.

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1 Introduction and scope

The preceding papers in this series moved from periodic prime-sieve words to centered two-factor kernels. Complete finite correlations recover the local Hardy–Littlewood product, character decomposition identifies the exact local modes, classical incomplete-character estimates transfer those modes to power-short boxes, and a multiplicative-large-sieve argument permits von Mangoldt weights on the two factor coordinates [15–19]. The last result counts prime factors p, q for which $pq + h$ avoids a growing finite set of primes. It does not make $pq + h$ prime.

Throughout, $\Lambda(t)$ and $\mathbf{1}_{\mathbb{P}}(t)$ are extended by zero for nonpositive integers t.

This paper studies that missing target-prime interface. There are two different transitions, and conflating them obscures the parity problem:

T1. replace the local target indicator $\mathbf{1}_{(mn+h,Q)=1}$ by $\Lambda(mn + h)$;

T2. connect a two-factor target-prime form back to the additive correlation $\sum_n \Lambda(n)\Lambda(n+h)$.

The first transition is a kernel and bilinear-form comparison. The second requires an exact convolution ledger and estimates for the resulting bilinear packets. Neither follows from a topological conjugacy, an ordinary spectral limit, or the finite character expansion alone.

1.1 The exact rough-to-prime defect

Let Q be odd and squarefree, put

$$G = (\mathbb{Z}/Q\mathbb{Z})^\times, \qquad \delta_Q = \frac{\varphi(Q)}{Q}, \qquad \kappa_Q = \prod_{p|Q} \frac{p-2}{p-1},$$

and suppose $(h, Q) = 1$. On factor coordinates which are Q-units, define

$$K_{Q,h}(m, n) = \mathbf{1}_{(mn+h,Q)=1}, \qquad K^\circ_{Q,h} = \kappa_Q^{-1} K_{Q,h} - 1.$$

For finite coefficient sequences $a = (a_m)$ and $b = (b_n)$ write

$$A_0 = \sum_m a_m, \qquad B_0 = \sum_n b_n, \tag{1}$$

$$\mathcal{B}_{Q,h}(a, b) = \sum_{m,n} a_m b_n K^\circ_{Q,h}(m, n), \tag{2}$$

$$\mathcal{P}_h(a, b) = \sum_{m,n} a_m b_n \Lambda(mn + h). \tag{3}$$

The pointwise defect

$$D_{Q,h}(m, n) = \Lambda(mn + h) - \delta_Q^{-1} K_{Q,h}(m, n) \tag{4}$$

gives the exact identity

$$\mathcal{P}_h(a, b) = \frac{\kappa_Q}{\delta_Q}\{A_0 B_0 + \mathcal{B}_{Q,h}(a, b)\} + \mathcal{D}_{Q,h}(a, b), \qquad \mathcal{D}_{Q,h}(a, b) = \sum_{m,n} a_m b_n D_{Q,h}(m, n). \tag{5}$$

Here

$$\frac{\kappa_Q}{\delta_Q} = \prod_{p|Q} \frac{p(p-2)}{(p-1)^2}. \tag{6}$$

Thus the preceding centered-kernel estimates control the first brace in (5); they do not estimate $\mathcal{D}_{Q,h}$. The identity is useful precisely because it prevents a local roughness theorem from being silently relabeled as a target-prime theorem.

1.2 Main results

The paper records the following exact identities, standard analytic consequences, and finite-information obstructions. No new level of distribution or parity-breaking estimate is claimed; the contribution is their joint formulation within the finite-dynamics framework.

R1. We resolve the finite truncation

$$\Lambda_{\le R}(N) = -\sum_{\substack{d\mid N\\ d\le R}} \mu(d)\log d$$

into exact congruence layers. For $(d,h)=1$, the layer $mn \equiv -h \pmod d$ is the graph of the involution $x \mapsto -hx^{-1}$ on $(\mathbb{Z}/d\mathbb{Z})^\times$. We give its complete Dirichlet-character expansion and show that its spectrum is contained in $\{+1,-1\}$.

R2. This comparison locates a structural loss. The actual squarefree d-layer has norm $\log d/\varphi(d)$, and the absolute norms over the prime-modulus subfamily contribute $\sim \log R$. Because different layers act on different spaces, this is not the divergence of a single operator. It says precisely that the summable scalar majorant used for the centered rough kernel cannot be transplanted verbatim; cancellation between moduli is neither excluded nor estimated.

R3. We prove a one-sided Bombieri–Vinogradov Type I theorem for $\sum_m \alpha_m \sum_n \Lambda(mn+h)$, uniformly for $|\alpha_m| \le 1$ when the modulus variable lies in the classical range $m \le x^{1/2}(\log x)^{-B}$. This is a direct consequence of the classical theorem, not a new level of distribution.

R4. Applying the same input to the exact Möbius–logarithm expansion of $\sum_n \Lambda(n)\Lambda(n+h)W(n/X)$ evaluates all coprime source divisors through $X^{1/2}(\log X)^{-B}$. The main term of this signed truncation equals the full formal pair singular-series constant. The Hardy–Littlewood asymptotic is then equivalent, by the exact divisor ledger, to cancellation of the complementary large-divisor Möbius–prime tail; that tail may still be of order X.

R5. We prove that a nonnegative observable periodic in the two factor coordinates cannot be a nonzero pointwise minorant for target primality on all sufficiently large pairs. We also give a nullspace theorem: modulo any fixed odd Q, two nonnegative point masses can have identical finite CRT, character, and centered-kernel data (the last at $\mathrm{rad}(Q)$) while their targets have opposite primality and Liouville values.

R6. At $Q = 5\cdot 7\cdot 11$ we supply a deterministic integer certificate. The locally indistinguishable targets are the square 148996 and the prime 149381; the latter is accompanied by a Lucas certificate.

The positive analytic result in R3 is deliberately one-sided. It can place a prime weight on one factor and on the target in an unbalanced range, but it does not permit an arbitrary second factor coefficient. In particular, the two true statements

$$\sum_{m,n} \Lambda(m)\Lambda(n)K_{Q,h}(m,n) \quad \text{and} \quad \sum_{m,n} \Lambda(m)\Lambda(mn+h)$$

cannot be intersected to infer a theorem for

$$\sum_{m,n} \Lambda(m)\Lambda(n)\Lambda(mn+h). \tag{7}$$

The missing covariance is displayed explicitly in section 3.

1.3 Position relative to known results

The Bombieri–Vinogradov input used below is classical [2, 10, 14]. Vaughan's identity is used only as an exact ledger; a generalized Heath–Brown identity would produce more multilinear packets but would not estimate them [9, 13]. The asymptotic sieve of Friedlander and Iwaniec [7] is an important positive comparison: in its axiomatic setting, additional Möbius-weighted bilinear information can break the parity barrier. The recent general framework of Ford and Maynard [5] likewise quantifies Type I/II requirements within its stated class of nonnegative sequences. We do not turn either framework into a theorem for (7).

There are also stronger theorems for differently averaged objects. Uniform Titchmarsh divisor results evaluate complete unweighted hyperbolic sums such as $\sum_{mn+h\le X} \Lambda(mn+h)$ with strong errors [1, Theorem 1.1]; see also Fouvry [6]. They do not provide a balanced dyadic rectangle estimate with arbitrary factor coefficients. Chen's original theorem represents every sufficiently large even integer as a prime plus a P_2-number [3]. A generalized fixed-linear-form theorem [8, Theorem 25.11] implies, for each fixed admissible even h, a lower bound for primes p with $p-h \in P_2$. Here P_2 means at most two prime factors and therefore includes the prime case; this theorem does not isolate $p = qr + h$ with q, r both prime. These are neighboring achievements, not consequences of our finite identities.

What is not claimed

We do not estimate the large-divisor tail isolated below, prove (7), obtain a lower bound for twin primes, or prove a new Bombieri–Vinogradov, dispersion, or Type II theorem. The finite nullspace and periodic-minorant results concern fixed local resolution. They do not exclude growing moduli, bilinear information, automorphic input, additive-combinatorial methods, or any other nonlocal argument. Conversely, the exact rewriting of an open correlation as one remainder is not presented as an independently easier conjecture.

2 Finite-resolution target-prime operators

This section gives two exact comparisons. The first is the rough-to-prime defect (5). The second resolves a truncated von Mangoldt target into arithmetic involutions. No asymptotic estimate is used.

2.1 The rough-to-prime bridge

Continue with the notation of section 1. Coefficients are always finite and are extended by zero away from their stated support.

Proposition 2.1 (Exact rough-to-prime defect)**.** *Let Q be odd and squarefree, let $(h, Q) = 1$, and suppose a_m, b_n are supported on Q-units. Then (5) holds exactly. In particular, no bound for $\mathcal{B}_{Q,h}(a, b)$ alone implies a bound for the prime-target form unless $\mathcal{D}_{Q,h}(a, b)$ is estimated independently.*

Proof. The definition of the centered kernel gives

$$K_{Q,h} = \kappa_Q(1 + K^{\circ}_{Q,h}).$$

After multiplication by $a_m b_n$ and summation,

$$\sum_{m,n} a_m b_n K_{Q,h}(m, n) = \kappa_Q\{A_0 B_0 + \mathcal{B}_{Q,h}(a, b)\}.$$

Now add the definition $\Lambda(mn + h) = \delta_Q^{-1} K_{Q,h}(m, n) + D_{Q,h}(m, n)$. The Euler product in (6) follows directly from the definitions of δ_Q and κ_Q. □

Remark 2.2 (Two dangerous specializations). At $h = 2$, take $Q = 1$, $a = \delta_1$, and $b_n = \Lambda(n)\mathbf{1}_{n\le X}$. Then

$$\mathcal{P}_2(a, b) = \sum_{n\le X} \Lambda(n)\Lambda(n + 2).$$

Thus a coefficient-uniform estimate strong enough to make the specialized defect $o(X)$, with the corresponding bridge main term retained, would give a positive linear-order asymptotic (indeed $\sim X$ in this $Q = 1$ normalization), and hence, after removing the $o(X)$ contribution of higher prime powers, infinitely many twin primes. At growing factor scales, finite dyadic choices $a_m = \Lambda(m)\mathbf{1}_{m\in I}$ and $b_n = \Lambda(n)\mathbf{1}_{n\in J}$ ask for three prime conditions $m, n, mn + h$; this is not the additive twin-prime correlation, but it remains a deep prime-producing problem. Neither specialization is supplied here.

2.2 Divisor resolution of the target weight

For $N \ge 1$ and $R \ge 1$, define

$$\Lambda_{\le R}(N) = -\sum_{\substack{d\mid N\\ d\le R}} \mu(d)\log d, \qquad \Delta_R(N) = \Lambda(N) - \Lambda_{\le R}(N). \tag{8}$$

The full identity $\Lambda = -\mu\log * \mathbf{1}$ gives

$$\Delta_R(N) = -\sum_{\substack{d\mid N\\ d>R}} \mu(d)\log d. \tag{9}$$

Thus Δ_R is an exact unresolved tail, not a probabilistic error term.

Let $I, J \subset \mathbb{Z}$ be finite, and let $a = (a_m)_{m\in I}$ and $b = (b_n)_{n\in J}$. We assume

$$a_m b_n \ne 0 \quad\Longrightarrow\quad (mn, h) = 1 \quad\text{and}\quad mn + h \ge 2. \tag{10}$$

This includes $h = 2$ with both factor coordinates supported on odd integers. Define

$$\mathcal{T}_{h,R}(a, b) = \sum_{m\in I}\sum_{n\in J} a_m b_n \Lambda_{\le R}(mn + h), \tag{11}$$

$$\mathcal{D}^{\mathrm{div}}_{h,R}(a, b) = \sum_{m\in I}\sum_{n\in J} a_m b_n \Delta_R(mn + h). \tag{12}$$

Then $\mathcal{P}_h = \mathcal{T}_{h,R} + \mathcal{D}^{\mathrm{div}}_{h,R}$ exactly.

For a modulus d and a Dirichlet character modulo d, extended by zero off the units, put

$$A_d(\chi) = \sum_{m\in I} a_m\chi(m), \qquad B_d(\chi) = \sum_{n\in J} b_n\chi(n). \tag{13}$$

Theorem 2.3 (Exact target-resolution expansion)**.** *Under* (10)*,*

$$\mathcal{T}_{h,R}(a, b) = -\sum_{\substack{d\le R\\ (d,h)=1}} \frac{\mu(d)\log d}{\varphi(d)} \sum_{\chi \bmod d} \overline{\chi(-h)} A_d(\chi)B_d(\chi). \tag{14}$$

Only squarefree d contribute. The complementary form is exactly

$$\mathcal{D}^{\mathrm{div}}_{h,R}(a, b) = -\sum_{m\in I}\sum_{n\in J} a_m b_n \sum_{\substack{d\mid mn+h\\ d>R}} \mu(d)\log d. \tag{15}$$

Proof. If $d \mid mn + h$ and (10) holds, then $(d, h) = 1$. Moreover m, n are units modulo d. Character orthogonality on $(\mathbb{Z}/d\mathbb{Z})^\times$ therefore gives

$$\mathbf{1}_{mn\equiv -h\,(d)} = \frac{1}{\varphi(d)} \sum_{\chi \bmod d} \chi(mn)\overline{\chi(-h)}. \tag{16}$$

Insert (16) in (8) and interchange finite sums. This proves (14). Equation (15) is (9) summed against the factor coefficients. □

2.3 Arithmetic involutions and the missing damping

For $(d, h) = 1$, let $G_d = (\mathbb{Z}/d\mathbb{Z})^\times$ with normalized counting measure and define

$$J_{h,d}(x) = -hx^{-1}, \qquad (U_{h,d}f)(x) = f(J_{h,d}(x)). \tag{17}$$

Proposition 2.4 (Involution spectrum)**.** *The map $J_{h,d}$ is a measure-preserving involution. Consequently $U_{h,d}$ is unitary, self-adjoint, and has spectrum contained in $\{+1, -1\}$. For every character $\chi \in \widehat{G}_d$,*

$$U_{h,d}\chi = \chi(-h)\overline{\chi}. \tag{18}$$

If $\chi = \overline{\chi}$, it is an eigenfunction with eigenvalue $\chi(-h)$. Every nonreal pair $\{\chi, \overline{\chi}\}$ spans one $+1$ and one -1 eigenline.

Proof. A direct calculation gives

$$J_{h,d}(J_{h,d}(x)) = -h(-hx^{-1})^{-1} = x.$$

Thus $J_{h,d}$ is a bijective involution and preserves normalized counting measure. Equation (18) follows from $\chi(x^{-1}) = \overline{\chi(x)}$. The remaining statements follow from $U_{h,d}^2 = I$ and the displayed action on each character pair. ☐

The normalized graph kernel of $U_{h,d}$ is

$$\Gamma_{h,d}(x, y) = \varphi(d)\mathbf{1}_{xy\equiv -h\,(d)} = \sum_{\chi \bmod d} \chi(x)\chi(y)\overline{\chi(-h)}. \tag{19}$$

Thus (14) is a weighted bundle of finite Koopman graph operators.

Corollary 2.5 (Layer norms and failure of an absolute majorant)**.** *Assume $h \neq 0$. For squarefree $d > 1$ with $(d, h) = 1$, the d-layer in the Möbius–logarithm resolution is*

$$V_{h,d} = -\frac{\mu(d)\log d}{\varphi(d)} U_{h,d}$$

and has operator norm

$$\|V_{h,d}\|_{2\to 2} = \frac{\log d}{\varphi(d)}. \tag{20}$$

Nevertheless the scalar absolute norms have no summable tail. Already

$$\sum_{\substack{p\le R\\ p\nmid h}} \|V_{h,p}\|_{2\to 2} = \sum_{\substack{p\le R\\ p\nmid h}} \frac{\log p}{p-1} = \log R + O_h(1). \tag{21}$$

Proof. For squarefree d, $|\mu(d)| = 1$, and proposition 2.4 gives $\|U_{h,d}\|_{2\to 2} = 1$. The last assertion follows from the prime number theorem and partial summation; replacing p by $p - 1$ changes the sum by a convergent amount. ☐

Because the operators $V_{h,d}$ act on different spaces $L^2(G_d)$, (21) is not the norm divergence of a single operator or of an unweighted direct-sum operator. It shows only that the scalar layer norms do not furnish an absolutely summable majorant. Hence the previous proof based on summing conductor norms cannot be transplanted verbatim; cancellation between moduli is neither excluded nor estimated here.

For the centered rough kernel of the preceding papers, a character of exact conductor q instead carries the multiplier

$$d_Q(\chi) = \prod_{p\mid q} \frac{1}{p-2}, \tag{22}$$

which is multiplied by $1/(p-2)$ whenever a new prime $p \geq 5$ enters the exact conductor (and is unchanged by the factor $p = 3$) [16, 18]. Equations (20)–(21) and (22) live in their respective normalizations, but the contrast is exact: the target-prime layers lack an absolutely summable scalar norm majorant, whereas the companion rough operator has a summable exact conductor-mass tail in its growing prime window. Thus the old absolute-majorant proof does not by itself control the large-divisor tail in (15).

3 The Bombieri–Vinogradov small-divisor interface

This section separates three statements which are easy to conflate. First, expanding the first von Mangoldt weight gives an exact divisor ledger; a *source divisor* means a divisor introduced by that expansion. Second, the Bombieri–Vinogradov theorem evaluates the part with $d \leq X^{1/2}(\log X)^{-B}$. Third, the complementary large-divisor tail is still unestimated. We also record a different one-sided application of Bombieri–Vinogradov in which one factor and the target are prime-weighted. None of these statements supplies a lower bound for a fixed prime pair.

Throughout the first part of the section, $h \neq 0$ is a fixed even integer, $W \in C_c^1((0,\infty))$ is supported in $[1,2]$, and

$$\mathcal{I}_W = \int_0^\infty W(u)\,du. \tag{23}$$

The restriction to even h is the admissible two-form case. The same identities hold for odd h, but then the local singular series vanishes.

3.1 An exact source-divisor ledger

Consider the smoothed fixed-shift correlation

$$\mathcal{C}_h(X;W) = \sum_{n\geq 1} \Lambda(n)\Lambda(n+h)W(n/X), \tag{24}$$

where X is sufficiently large that every target in the support is positive. The elementary identity

$$\Lambda(n) = -\sum_{d\mid n} \mu(d)\log d \tag{25}$$

gives the following finite decomposition.

Proposition 3.1 (Exact source Möbius–log ledger)**.** *For every X and every $1 \leq D \leq 2X$,*

$$\begin{aligned}\mathcal{C}_h(X;W) &= -\sum_{d\leq 2X} \mu(d)\log d \sum_{m\geq 1} \Lambda(dm+h)W(dm/X) &(26)\\ &= \mathcal{C}_{h,\leq D}(X;W) + \mathcal{T}_{h,>D}(X;W), &(27)\end{aligned}$$

where

$$\begin{aligned}\mathcal{C}_{h,\leq D}(X;W) &= -\sum_{d\leq D} \mu(d)\log d \sum_{m\geq 1} \Lambda(dm+h)W(dm/X), &(28)\\ \mathcal{T}_{h,>D}(X;W) &= -\sum_{D<d\leq 2X} \mu(d)\log d \sum_{m\geq 1} \Lambda(dm+h)W(dm/X). &(29)\end{aligned}$$

All these identities are exact.

Proof. Insert (25) into (24), interchange two finite sums, and write $n = dm$. Since $W(n/X)$ vanishes unless $X \leq n \leq 2X$, no divisor larger than $2X$ occurs. Splitting at D proves (27). □

Divisors not coprime to h do not form the analytic obstruction. They are supported on a finite collection of prime-power targets.

Lemma 3.2 (The non-coprime source part)**.** *For every $\varepsilon > 0$,*

$$\sum_{\substack{d\le 2X\\(d,h)>1}} |\mu(d)|\log(2d)\sum_{m\ge 1}\Lambda(dm+h)|W(dm/X)| \ll_{h,W,\varepsilon} X^{\varepsilon}. \tag{30}$$

The same bound holds after imposing either $d \le D$ or $d > D$.

Proof. After writing $n = dm$, the left side is at most

$$\sum_{n\asymp X}\Lambda(n+h)|W(n/X)|\sum_{\substack{d|n\\(d,h)>1}}|\mu(d)|\log(2d).$$

If the inner sum is nonempty, some prime $p \mid h$ divides n. Hence $p \mid n+h$. The condition $\Lambda(n+h) \neq 0$ then forces $n+h = p^k$. For each of the finitely many primes $p \mid h$, only $O_h(1)$ powers p^k lie in an interval of the form $[X+h, 2X+h]$. Finally,

$$\sum_{d|n}|\mu(d)|\log(2d) \ll \tau(n)\log(2n) \ll_{\varepsilon} X^{\varepsilon},$$

after reducing ε if necessary. This proves the claim. □

3.2 The one-sided Bombieri–Vinogradov input

We use the maximal form of the Bombieri–Vinogradov theorem. This is a classical input, not a new distribution theorem [2, 10, 14].

Theorem 3.3 (One-sided smooth Type I estimate)**.** *Fix $h \neq 0$, $A > 0$, and $W \in C_c^1((0,\infty))$. There is a constant $B = B(A,W) > 0$ with the following properties.*

Let $x = MN$ be sufficiently large, let $M, N \ge 2$, and suppose

$$2M \le \frac{x^{1/2}}{(\log x)^B}. \tag{31}$$

For arbitrary complex coefficients $|\alpha_m| \le 1$ supported on $M < m \le 2M$ and $(m,h) = 1$,

$$\sum_{M<m\le 2M}\alpha_m\left\{\sum_{n\ge 1}\Lambda(mn+h)W(n/N) - \frac{mN}{\varphi(m)}\mathcal{I}_W\right\} \ll_{A,h,W} \frac{x}{(\log x)^A}. \tag{32}$$

In particular, the main term in the mth row is $mN\mathcal{I}_W/\varphi(m)$, not $N\mathcal{I}_W$.

There is also a fixed-source-scale form. If X is sufficiently large,

$$D \le \frac{X^{1/2}}{(\log X)^B} \tag{33}$$

and $|\alpha_d| \le 1$ is supported on $d \le D$ and $(d,h) = 1$, then

$$\sum_{\substack{d\le D\\(d,h)=1}}\alpha_d\left\{\sum_{m\ge 1}\Lambda(dm+h)W(dm/X) - \frac{X}{\varphi(d)}\mathcal{I}_W\right\} \ll_{A,h,W} \frac{X}{(\log X)^A}. \tag{34}$$

Proof. Put

$$E(y;q,a) = \sum_{\substack{r\le y\\ r\equiv a \pmod q}} \Lambda(r) - \frac{y}{\varphi(q)} \qquad ((a,q)=1).$$

The maximal Bombieri–Vinogradov theorem states that, after increasing B in terms of A, the sum over q up to the right side of (31) of $\max_{(a,q)=1}\max_{y\le Cx}|E(y;q,a)|$ is $O_A(x(\log x)^{-A})$; here C depends only on the support of W.

For fixed m, the substitution $r = mn + h$ places the target in the reduced class $h \pmod m$. Stieltjes partial summation against $W((r-h)/(mN))$ shows that the expression in braces in (32) is bounded by a constant depending on W times

$$\max_{y\le Cx+|h|} |E(y;m,h)|.$$

The integral of the continuous main term is

$$\frac{1}{\varphi(m)}\int_0^\infty W((t-h)/(mN))\,dt = \frac{mN}{\varphi(m)}\mathcal{I}_W.$$

Summing the error with $|\alpha_m|\le 1$ proves (32). The same argument with the weight $W((r-h)/X)$ gives (34); in that case the continuous main term is $X\mathcal{I}_W/\varphi(d)$. ☐

The coefficient $\mu(d)\log d$ in the exact source ledger costs only one additional logarithm in theorem 3.3. The main terms are summed by the following standard singular-series coefficient identity.

3.3 The singular coefficient and small-divisor evaluation

For even $h \ne 0$, define

$$\mathfrak{S}(h) = 2\prod_{p>2}\left(1-\frac{1}{(p-1)^2}\right)\prod_{\substack{p\mid h\\ p>2}}\frac{p-1}{p-2}. \tag{35}$$

Lemma 3.4 (Singular-series coefficient)**.** *For every fixed even $h\ne 0$ and every $A>0$,*

$$-\sum_{\substack{d\le z\\ (d,h)=1}}\frac{\mu(d)\log d}{\varphi(d)} = \mathfrak{S}(h) + O_{A,h}((\log z)^{-A}) \qquad (z\ge 3). \tag{36}$$

Proof. For $\Re s > 0$, set

$$F_h(s) = \sum_{\substack{d\ge 1\\ (d,h)=1}}\frac{\mu(d)}{\varphi(d)d^s} = \prod_{p\nmid h}\left(1-\frac{p^{-s}}{p-1}\right) \tag{37}$$

$$= \frac{G_h(s)}{\zeta(1+s)}, \tag{38}$$

where

$$G_h(s) = \prod_{p\mid h}(1-p^{-1-s})^{-1}\prod_{p\nmid h}\frac{1-p^{-s}/(p-1)}{1-p^{-1-s}}. \tag{39}$$

The second product converges absolutely in a fixed half-plane containing $s=0$. At zero its local factors give

$$G_h(0) = 2\prod_{\substack{p>2\\ p\nmid h}}\frac{p(p-2)}{(p-1)^2}\prod_{\substack{p\mid h\\ p>2}}\frac{p}{p-1} = \mathfrak{S}(h).$$

We now justify the truncation quantitatively rather than differentiate at the boundary of convergence. Write

$$G_h(s) = \sum_{r\geq 1} \frac{g_h(r)}{r^s}.$$

The coefficients are multiplicative and, for $\nu \geq 1$, satisfy

$$g_h(p^\nu) = \begin{cases} p^{-\nu}, & p \mid h, \\ -\{p^\nu(p-1)\}^{-1}, & p \nmid h. \end{cases}$$

Hence, for any fixed $0 < \eta < 1$, the Euler product of the absolute local sums converges: away from the finitely many primes dividing h, its nonconstant term is $O_\eta(p^{-2+\eta})$. In particular,

$$\sum_{r\geq 1} |g_h(r)| r^\eta < \infty. \tag{40}$$

Put

$$A_0(y) = \sum_{k\leq y} \frac{\mu(k)}{k}, \qquad A_1(y) = \sum_{k\leq y} \frac{\mu(k)\log k}{k}.$$

The classical zero-free region and partial summation imply, for every $C > 0$ and $y \geq 1$,

$$A_0(y) \ll_C (\log(2y))^{-C}, \qquad A_1(y) = -1 + O_C((\log(2y))^{-C}); \tag{41}$$

see Iwaniec and Kowalski [10], Montgomery and Vaughan [11]. In the common half-plane of absolute convergence, (38) gives the coefficient identity

$$\frac{\mu(d)\mathbf{1}_{(d,h)=1}}{\varphi(d)} = \sum_{rk=d} g_h(r) \frac{\mu(k)}{k}.$$

Consequently

$$-\sum_{\substack{d\leq z \\ (d,h)=1}} \frac{\mu(d)\log d}{\varphi(d)} = -\sum_{r\leq z} g_h(r) \left\{(\log r) A_0(z/r) + A_1(z/r)\right\}. \tag{42}$$

For $r \leq z^{1/2}$, (41), with C chosen larger than $A+2$, makes the difference between the braces and -1 equal to $O_{A,h}((1+\log r)(\log z)^{-A-1})$. Its sum is admissible by (40). For $r > z^{1/2}$, the elementary bounds $|A_0(y)| \ll \log(2y)$ and $|A_1(y)| \ll (\log(2y))^2$ combine with (40) to give $O_{A,h}((\log z)^{-A})$. The omitted tail $\sum_{r>z} g_h(r)$ has the same bound. Thus (42) equals $G_h(0) + O_{A,h}((\log z)^{-A})$, and $G_h(0) = \mathfrak{S}(h)$ proves the lemma. □

Corollary 3.5 (Evaluation of the small source-divisor range)**.** *Fix $A > 0$. There is $B = B(A, h, W) > 0$ such that, for all sufficiently large X, with*

$$D = \left\lfloor \frac{X^{1/2}}{(\log X)^B} \right\rfloor, \tag{43}$$

one has

$$\mathcal{C}_{h,\leq D}(X; W) = \mathfrak{S}(h) X \mathcal{I}_W + O_{A,h,W}\left(\frac{X}{(\log X)^A}\right). \tag{44}$$

Proof. By lemma 3.2, the terms with $(d,h) > 1$ contribute $O_{h,W,\varepsilon}(X^\varepsilon)$. On the coprime terms, apply (34) with

$$\alpha_d = -\frac{\mu(d)\log d}{\log D}.$$

After multiplying back by $\log D$ and choosing the exponent in the Bombieri–Vinogradov error larger than $A+2$, this gives

$$\mathcal{C}_{h,\le D}(X;W) = -X\mathcal{I}_W \sum_{\substack{d\le D\\(d,h)=1}} \frac{\mu(d)\log d}{\varphi(d)} + O_{A,h,W}(X(\log X)^{-A}).$$

Now use lemma 3.4. Since $\log D \asymp \log X$, its truncation error is absorbed into the displayed error. Taking, for example, $\varepsilon = 1/2$ absorbs the non-coprime part as well. □

The content of corollary 3.5 is precisely limited to the short source-divisor range. Combining it with the exact split gives an equivalence, not a solution.

Theorem 3.6 (Exact large-divisor-tail equivalence)**.** *For each $A>0$, choose $B = B(A,h,W)$ as in corollary 3.5, and let D be given by (43). Then, for all sufficiently large X,*

$$\mathcal{C}_h(X;W) - \mathfrak{S}(h)X\mathcal{I}_W = \mathcal{T}_{h,>D}(X;W) + O_{A,h,W}\left(\frac{X}{(\log X)^A}\right). \tag{45}$$

Consequently,

$$\mathcal{C}_h(X;W) \sim \mathfrak{S}(h)X\mathcal{I}_W \quad\Longleftrightarrow\quad \mathcal{T}_{h,>D}(X;W) = o(X), \tag{46}$$

provided $\mathcal{I}_W \neq 0$.

Proof. Substitute (44) into the exact identity (27). This proves (45), and the equivalence follows. □

Remark 3.7 (What the truncation evaluation does not say)*.* The right side of (45) contains the divisors $d>D$, where the direct arithmetic-progression argument has no Bombieri–Vinogradov coverage. The assertion that this tail is $o(X)$ is exactly the missing fixed-shift cancellation. Naming the tail, or noting that its coefficients have signs, does not estimate it. In particular, theorem 3.6 is not a reduction of the twin-prime problem to an independently easier theorem.

3.4 A prime-factor/target-prime marginal

The same one-sided theorem gives a standard but distinct prime-target asymptotic. Let $W_1, W_2 \in C_c^1((0,\infty))$ be supported in $[1,2]$, put $x = MN$, and define

$$\mathcal{M}_h(M,N) = \sum_{m,n\ge 1} \Lambda(m)\Lambda(mn+h)W_1(m/M)W_2(n/N), \tag{47}$$

$$\mathcal{P}_h(M,N) = \sum_{m,n\ge 1} \Lambda(m)\Lambda(n)\Lambda(mn+h)W_1(m/M)W_2(n/N), \tag{48}$$

$$\mathcal{V}_h(M,N) = \sum_{m,n\ge 1} \Lambda(m)(\Lambda(n)-1)\Lambda(mn+h)W_1(m/M)W_2(n/N). \tag{49}$$

Proposition 3.8 (Marginal theorem and covariance ledger)**.** *Suppose $M,N\to\infty$ and, for a sufficiently large B,*

$$2M \le \frac{x^{1/2}}{(\log x)^B}. \tag{50}$$

Then

$$\mathcal{M}_h(M,N) = MN\left(\int_0^\infty W_1\right)\left(\int_0^\infty W_2\right) + o(MN). \tag{51}$$

Moreover, the identity

$$\mathcal{P}_h(M,N) = \mathcal{M}_h(M,N) + \mathcal{V}_h(M,N) \tag{52}$$

is exact. Hence the expected three-prime asymptotic, written in a form that remains meaningful when a weight integral vanishes,

$$\mathcal{P}_h(M,N) = \mathfrak{S}(h)MN\left(\int_0^\infty W_1\right)\left(\int_0^\infty W_2\right) + o(MN), \tag{53}$$

is equivalent, in this range, to

$$\mathcal{V}_h(M,N) = (\mathfrak{S}(h)-1)MN\left(\int_0^\infty W_1\right)\left(\int_0^\infty W_2\right) + o(MN). \tag{54}$$

In particular, when $(\mathfrak{S}(h)-1)(\int W_1)(\int W_2) \neq 0$*, the missing covariance is not predicted to be* $o(MN)$*.*

Proof. In (32), take

$$C_1 = \max\{1, \|W_1\|_\infty\}, \qquad \alpha_m = \frac{\Lambda(m)W_1(m/M)}{C_1 \log(3M)}$$

and then multiply the result by $C_1\log(3M)$. Thus $|\alpha_m| \le 1$ on the support in question. By choosing the logarithmic saving one power larger, the resulting error is $o(MN)$. Rows with $(m,h) > 1$ are negligible: if $\Lambda(m) \neq 0$ and a prime $p \mid (m,h)$, then m is a power of p, and $mn+h$ can carry von Mangoldt weight only when it too is a power of p. For each $p \mid h$, a dyadic interval contains $O_h(1)$ relevant powers $m = p^a$, and the target interval contains $O_h(1)$ powers $mn+h = p^b$; their total contribution is $O_h((\log x)^2) = o(MN)$.

The main term is

$$N\left(\int_0^\infty W_2\right)\sum_m \Lambda(m)W_1(m/M)\frac{m}{\varphi(m)}.$$

The prime number theorem and the negligible contribution of higher prime powers give

$$\sum_m \Lambda(m)W_1(m/M)\frac{m}{\varphi(m)} = M\int_0^\infty W_1(u)\,du + o(M),$$

which proves (51).

Equation (52) is obtained by writing $\Lambda(n) = 1 + (\Lambda(n)-1)$. For completeness, the local factor of the three conditions $m, n, mn+h$ is indeed (35). If $p \nmid h$, there are $(p-1)(p-2)$ pairs $(a,b) \in \mathbb{F}_p^2$ for which a, b, and $ab+h$ are all nonzero. Relative to three independent prime conditions, the normalized factor is

$$\frac{(p-1)(p-2)/p^2}{(1-1/p)^3} = \frac{p(p-2)}{(p-1)^2}.$$

If $p \mid h$, there are $(p-1)^2$ admissible pairs and the factor is $p/(p-1)$. Their product is $\mathfrak{S}(h)$. Subtracting the unconditional marginal asymptotic from (53) proves the equivalence with (54). ☐

Remark 3.9 (Two marginals do not determine their intersection)*.* The marginal theorem requires m and $mn+h$ to be prime-weighted but leaves n unrestricted. The preceding local-kernel results require both factor variables to be prime-weighted but leave the target only rough. These two statements cannot be intersected without estimating (49). Whenever its coefficient is nonzero, the main term predicted in (54) quantifies that missing dependence.

3.5 A compact target-side Vaughan ledger

For later use, we finish with an exact target-side decomposition. It is a standard form of Vaughan's identity [10, 13]; generalized identities produce more multilinear packets but do not estimate them by themselves [9].

Proposition 3.10 (Exact Vaughan ledger)**.** *Let F be any finitely supported function on the positive integers and let $U, V \geq 1$. Put*

$$\beta_V(k) = \sum_{\substack{d|k \\ d\leq V}} \mu(d). \tag{55}$$

Then

$$\begin{aligned}\sum_{t} \Lambda(t)F(t) = \sum_{t\leq U} \Lambda(t)F(t) - \sum_{\substack{\ell\leq U \\ d\leq V}} \Lambda(\ell)\mu(d) \sum_{r\geq 1} F(\ell dr) \\ + \sum_{d\leq V} \mu(d) \sum_{r\geq 1} (\log r)F(dr) - \sum_{\substack{\ell>U \\ k>1}} \Lambda(\ell)\beta_V(k)F(\ell k).\end{aligned} \tag{56}$$

Moreover, $\beta_V(k) = 0$ for $1 < k \leq V$.

Proof. Write $\mu_{\leq V}(d) = \mu(d)\mathbf{1}_{d\leq V}$, so that $\beta_V = \mu_{\leq V} * \mathbf{1}$. The second term in (56) is $-(\Lambda_{\leq U} * \beta_V)(t)$, while the third is $(\mu_{\leq V} * \log)(t)$. The last term equals

$$-(\Lambda_{>U} * \beta_V)(t) + \Lambda_{>U}(t),$$

because the omitted value $k = 1$ has $\beta_V(1) = 1$. Adding the first term therefore gives $\Lambda(t)$ plus

$$(\mu_{\leq V} * \log)(t) - (\Lambda * \mu_{\leq V} * \mathbf{1})(t),$$

which is zero because $\Lambda * \mathbf{1} = \log$. Finally, if $1 < k \leq V$, then every divisor of k is at most V, and $\beta_V(k) = \sum_{d|k} \mu(d) = 0$. □

To see the remaining geometry, take

$$F_h^{a,b}(t) = \sum_{m,n} a_m b_n \mathbf{1}_{mn+h=t}. \tag{57}$$

Here the coefficients are finite and satisfy $a_m b_n \neq 0 \Rightarrow mn + h \geq 1$. The middle two packets in (56) become the Type I congruence sums

$$-\sum_{\substack{\ell\leq U \\ d\leq V}} \Lambda(\ell)\mu(d) \sum_{mn\equiv -h \,(\mathrm{mod}\, \ell d)} a_m b_n, \tag{58}$$

$$\sum_{d\leq V} \mu(d) \sum_{mn\equiv -h \,(\mathrm{mod}\, d)} a_m b_n \log\left(\frac{mn+h}{d}\right), \tag{59}$$

whereas the last packet is

$$-\sum_{\substack{\ell>U \\ k>V}} \Lambda(\ell)\beta_V(k) \sum_{mn+h=\ell k} a_m b_n. \tag{60}$$

Thus the unresolved fixed-shift object is supported on the shifted multiplicative surface

$$\ell k - mn = h, \qquad \ell > U, \quad k > V. \tag{61}$$

The finite roughness kernel tests congruence avoidance by prescribed primes; it does not estimate (60). Consequently the exact Vaughan ledger identifies a Type II interface but does not close it.

4 Finite-information barriers

The exact character formulas above describe everything visible at a fixed finite congruence resolution. This section proves two correspondingly finite limitations. The first concerns nonnegative pointwise minorants. The second is a nullspace statement for all observables that factor through a fixed CRT pushforward.

4.1 Periodic prime minorants collapse

Call $F : \mathbb{Z}^2 \to \mathbb{R}$ *Q-local* if

$$F(m + Q, n) = F(m, n + Q) = F(m, n) \tag{62}$$

for all m, n. Such an observable is a function on $(\mathbb{Z}/Q\mathbb{Z})^2$. Every fixed-$Q$ factor-residue statistic, finite character moment whose modulus divides Q, and target-rough kernel at a squarefree modulus dividing Q is Q-local.

Theorem 4.1 (Periodic-minorant collapse)**.** *Fix integers $Q \geq 1$ and $h \neq 0$. Let $F : \mathbb{Z}^2 \to [0, \infty)$ be Q-local. If, for all sufficiently large positive m, n,*

$$F(m, n) \leq \mathbf{1}_{\mathbb{P}}(mn + h), \tag{63}$$

then $F \equiv 0$. The same conclusion holds if the right-hand side is $\Lambda(mn + h)$.

Proof. Fix any residue pair $(a, b) \pmod Q$. Choose distinct primes r, s with $(rs, Qh) = 1$. The Chinese remainder theorem gives simultaneous solutions

$$m \equiv a \pmod Q, \quad n \equiv b \pmod Q, \quad m \equiv 1 \pmod{rs}, \quad n \equiv -h \pmod{rs}.$$

Adding multiples of Qrs makes m, n arbitrarily large and ensures $mn + h > rs$. The target is divisible by the two distinct primes r, s; hence it is composite and is not a prime power. Therefore both $\mathbf{1}_{\mathbb{P}}(mn + h)$ and $\Lambda(mn + h)$ are zero. Periodicity and nonnegativity give

$$0 \leq F(a, b) = F(m, n) \leq 0.$$

Since (a, b) was arbitrary, F vanishes identically. □

Remark 4.2 (Precise scope)*.* The theorem concerns a fixed period and a pointwise nonnegative minorant. It does not apply to signed identities, mean-square approximations, moduli growing with the observation scale, or bilinear and dispersion estimates. It says only that a fixed finite-state local observable cannot itself be the positive target-prime detector required for a lower-bound sieve.

4.2 A CRT nullspace witness

For a finite signed measure ν on $\mathbb{N}^2$, let

$$\Pi_Q \nu = (\pi_Q)_{\#} \nu, \qquad \pi_Q(m, n) = (m \bmod Q, n \bmod Q). \tag{64}$$

Every Q-local linear statistic factors through Π_Q.

Define the target-prime and target-parity functionals

$$L_{\text{prime}}(\nu) = \int \mathbf{1}_{\mathbb{P}}(mn + 2)\, d\nu(m, n), \tag{65}$$

$$L_{\lambda}(\nu) = \int \lambda(mn + 2)\, d\nu(m, n), \tag{66}$$

where $\lambda(N) = (-1)^{\Omega(N)}$.

Theorem 4.3 (Fixed-CRT non-identifiability)**.** *Let $Q \geq 3$ be odd. There exist two nonnegative unit point masses $\nu_\square$ and ν_{p} on $\mathbb{N}^2$ such that*

$$\Pi_Q \nu_\square = \Pi_Q \nu_{\mathsf{p}}, \tag{67}$$

but

$$L_{\text{prime}}(\nu_\square) = 0, \qquad L_{\text{prime}}(\nu_{\mathsf{p}}) = 1, \tag{68}$$
$$L_\lambda(\nu_\square) = +1, \qquad L_\lambda(\nu_{\mathsf{p}}) = -1. \tag{69}$$

Both targets and both factor coordinates are units modulo Q. Consequently the two measures have identical values for every fixed-Q CRT observable, every Dirichlet-character observable modulo a divisor of Q on the factor residues, and the centered target-rough kernel at the squarefree modulus $\operatorname{rad}(Q)$.

Proof. Put
$$T_\square = (Q+1)^2, \qquad x_\square = (1, T_\square - 2).$$
Dirichlet's theorem supplies a prime $\ell \equiv 1 \pmod{Q}$, which may be chosen arbitrarily large [11, Chapter 6]. Put $x_{\mathsf{p}} = (1, \ell - 2)$ and let the two measures be the corresponding point masses. Since both targets are congruent to $1 \pmod{Q}$,

$$x_\square \equiv x_{\mathsf{p}} \equiv (1, -1) \pmod{Q}.$$

This proves (67) and the unit assertions. The first target is a square, so its Liouville value is $+1$ and it is composite; the second is prime, so its Liouville value is -1. This proves (68)–(69). Every stated local statistic is a function of $\pi_Q(m,n)$ and therefore has the same integral against the two measures. ☐

Let $v = \nu_\square - \nu_{\mathsf{p}}$. Then

$$\Pi_Q v = 0, \qquad L_{\text{prime}}(v) = -1, \qquad L_\lambda(v) = 2. \tag{70}$$

Thus neither target functional belongs to the row space generated by all fixed-Q local observables.

Corollary 4.4 (Minimax consequence)**.** *Any estimator of L_{prime} which depends only on $\Pi_Q \nu$ has worst-case absolute error at least $1/2$ on the two point masses in theorem 4.3. For L_λ, the corresponding lower bound is 1.*

Proof. The estimator must output the same number on two inputs with identical pushforward. A single real number cannot be within less than $1/2$ of both 0 and 1, nor within less than 1 of both -1 and $+1$. ☐

4.3 An exact certificate at $Q = 385$

Take
$$Q = 5 \cdot 7 \cdot 11 = 385, \qquad h = 2. \tag{71}$$

The two target values in the certificate are

$$T_\square = 386^2 = 148996 = 2^2 \cdot 193^2, \tag{72}$$
$$T_{\mathsf{p}} = 148996 + 385 = 149381. \tag{73}$$

They are represented by

$$x_\square = (1, 148994), \qquad x_{\mathsf{p}} = (1, 149379), \tag{74}$$

and both reduce to $(1, 384)$ modulo 385. Their common local signature is

$$
\begin{array}{cccc}
\hline
p & m \bmod p & n \bmod p & (mn+2) \bmod p \\
\hline
5 & 1 & 4 & 1 \\
7 & 1 & 6 & 1 \\
11 & 1 & 10 & 1 \\
\hline
\end{array}
\tag{75}
$$

Every window prime therefore treats both targets as the same survivor. The local density and centered-kernel value are exactly

$$\kappa_Q = \frac{3}{4}\frac{5}{6}\frac{9}{10} = \frac{9}{16}, \qquad K^{\circ}_{Q,2}(x_{\square}) = K^{\circ}_{Q,2}(x_{\mathsf{p}}) = \frac{7}{9}. \tag{76}$$

For completeness, primality of 149381 is certified without probabilistic testing. Its predecessor factors completely as

$$149381 - 1 = 149380 = 2^2 \cdot 5 \cdot 7 \cdot 11 \cdot 97. \tag{77}$$

With witness $a = 3$,

$$3^{149380} \equiv 1 \pmod{149381}, \tag{78}$$

and for every $q \in \{2, 5, 7, 11, 97\}$,

$$\gcd\big(3^{149380/q} - 1, 149381\big) = 1. \tag{79}$$

The residues of $3^{149380/q}$ modulo 149381, in the displayed order of q, are

$$149380, \quad 11332, \quad 34444, \quad 38519, \quad 28976. \tag{80}$$

The Lucas criterion and (77) prove that 149381 is prime. Explicitly, the criterion says that if $a^{N-1} \equiv 1 \pmod N$ and $\gcd(a^{(N-1)/q} - 1, N) = 1$ for every prime $q \mid N - 1$, then N is prime; (78)–(80) verify exactly these hypotheses.

The accompanying program recomputes all factorizations, congruences, fractions, functional gaps, and the Lucas certificate using only exact standard-library integer arithmetic. It also regenerates the retained JSON certificate byte for byte; see section 5.

Remark 4.5 (Relation to the classical parity problem). The square/prime witness has opposite Liouville parity and annihilates every fixed-Q observable, so it is an exact finite-information obstruction. It is not the classical sieve parity phenomenon. For Selberg's limitation and an explicit model sequence satisfying strong remainder axioms while containing no prime mass, see Friedlander and Iwaniec [7], Selberg [12]; for a modern limitation within a specified Selberg-sieve framework, see D. H. J. Polymath [4]. Those results compare much richer asymptotic data and specify which additional bilinear axioms can change the conclusion. Our theorem is intentionally narrower and is used only to delimit the finite dynamical information of the present series.

5 Certificates, comparison ledger, and boundaries

5.1 Exact reproducibility certificate

The directory `experiments/` contains a pure standard-library Python implementation of the finite witness in section 4. It uses integers and `Fraction` objects only; there is no floating-point arithmetic, random sampling, external primality library, or fitted parameter. The retained certificate is

`experiments/data/exact-certificate.json`.

Table 1: Exact finite witness retained in the certificate.

quantity	exact value
Q, h	$385, 2$
square target	$148996 = 2^2 \cdot 193^2$
prime target	149381
common factor residue	$(1, 384) \pmod{385}$
common target residue	$1 \pmod{385}$
κ_Q	$9/16$
common $K^{\circ}_{Q,2}$	$7/9$
$L_{\mathrm{prime}}(v)$	-1
$L_{\lambda}(v)$	2
prime minimax lower bound	$1/2$
Liouville minimax lower bound	1
Lucas witness	3

Its SHA-256 digest is

`0C6BDAC0FDBD5BAB3308AAA907F361D9903B5A88406F6F466B621490C9AB7004`.

The main exact entries are summarized in table 1.

Running

```
python exact_defect_certificate.py --output data/exact-certificate.json
python -m unittest -v test_exact_defect_certificate.py
```

recomputes the JSON object and verifies byte-for-byte equality with the tracked file. The tests cover the two factorizations, the complete Lucas certificate, CRT pushforward equality, all three prime-window signatures, the exact centered-kernel value, both functional gaps, and both minimax bounds.

Remark 5.1 (What the certificate certifies). The certificate is a proof aid for a finite algebraic witness. It is not numerical evidence that a growing-window defect is small or large, and no asymptotic theorem depends on extrapolating its values.

5.2 What is already known for neighboring sums

The target-prime form changes character when the coefficient geometry changes. Three nearby statements should not be confused.

K1. The complete unweighted hyperbolic sum can be rewritten as a shifted divisor problem:

$$\sum_{mn+h\le X} \Lambda(mn+h) = \sum_{k\le X} \Lambda(k)\tau(k-h)$$

up to the elementary boundary convention. Strong uniform asymptotics for this type of sum are known [1, Theorem 1.1]; see also Fouvry [6]. We make no novelty claim for it.

K2. The one-sided form with arbitrary α_m and no β_n is controlled by Bombieri–Vinogradov in the exact modulus range (31); this is theorem 3.3. Inserting a free second coefficient sequence destroys the direct arithmetic-progression reduction.

K3. A balanced dyadic form

$$\sum_{m\sim M}\sum_{n\sim N} \alpha_m\beta_n\Lambda(mn+h)$$

with broad coefficient classes is a genuinely different Type II or dispersion problem. No theorem in this paper estimates it at a prescribed shift.

Likewise, the generalized Chen theorem recorded in Friedlander and Iwaniec [8, Theorem 25.11] supplies many primes p for which $p-h$ has at most two prime factors, in an appropriate fixed-even-shift formulation. It does not allocate the mass between the prime and exact semiprime sectors. This is a classical near-miss, not a corollary of the CRT nullspace theorem.

5.3 A strict ledger of implications

The results proved here have the following one-way relationships:

$$\begin{aligned} \text{finite CRT kernel} \quad + \quad \text{TPC-5 conductor estimates} \quad &\Longrightarrow \quad \text{rough-target main term,} \\ \text{Bombieri–Vinogradov} \quad &\Longrightarrow \quad \text{one-sided target-prime Type I range,} \\ \text{full fixed-shift prime-pair asymptotic} \quad &\Longleftrightarrow \quad \text{large-divisor tail cancellation in theorem 3.6.} \end{aligned} \tag{81}$$

The first two lines do not combine automatically, because they control different weights and different coefficient norms. The last line is an exact equivalence, not a proof of either side.

The unresolved form produced by Vaughan's identity in section 3.5 has the multiplicative-surface geometry

$$\ell k - mn = h. \tag{82}$$

Finite target roughness only records whether the right side of $mn+h$ meets selected residue classes. It gives neither cancellation between the two factorizations in (82) nor positivity for a prime target.

5.4 Strict boundary of the claims

For clarity, none of the following is asserted.

B1. We do not prove that $mn+h$ is prime for any infinite family of factor pairs, and do not prove a lower bound for $\Lambda(m)\Lambda(n)\Lambda(mn+h)$.

B2. We do not estimate $\sum_n \Lambda(n)\Lambda(n+2)$ beyond the exact decomposition of theorem 3.6. The large-divisor tail is the unsolved content.

B3. The one-sided Bombieri–Vinogradov theorem has no arbitrary second coefficient sequence. Calling it a balanced Type II estimate would be false.

B4. Vaughan's and Heath–Brown's identities are algebraic decompositions. Merely naming their Type I/II packets supplies no estimate for those packets.

B5. The periodic-minorant and nullspace theorems concern a fixed finite period. They are not universal impossibility theorems for analytic number theory and do not rule out increasing resolution or nonlocal input.

B6. No spectral isomorphism between the logistic map and a prime-target operator is claimed. The dynamics used here consists of finite arithmetic shifts, congruence graph involutions, and their exact Fourier spectra.

5.5 Conclusion

Within the present framework, two elementary exact identities separate the finite rough term from the target von Mangoldt term and decompose a finite divisor truncation into congruence involutions. Classical Bombieri–Vinogradov theory evaluates the one-sided Type I range up to $X^{1/2}(\log X)^{-B}$. In the additive ledger, the standard main-term calculation for that signed truncation yields the full formal pair singular-series constant; the complementary fixed-shift tail remains unestimated.

The negative theorems explain why this remainder cannot be replaced by a fixed periodic prime minorant or inferred from finite CRT spectra. Within the present divisor/operator framework, a next advance would require a genuinely nonlocal estimate—for example a coefficient-stable Type II or dispersion theorem for a specified portion of (82). Any such theorem must state its coefficient class, scale, averaging variable, and fixed-shift quantifier explicitly.

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